



ICPC SOUTH PACIFIC PRELIMINARY CONTEST LEVEL B

SEPTEMBER 14, 2025

Contest Problems

A: Arr-Gee-Bee Rainbow
B: Broken Heart
C: Combinatorial Dice
D: Dictionary Shuffle
E: Euclidean Extension
F: Firefighting Robots
G: Grandma's House
H: Humanoid Robot Tango
I : I Get Extra Votes
J : Jeremy's Style Guide
K: Kayak is not a Palindrone
L: Love Me

Local Contest Sponsors



Multi-Regional Sponsor



Global ICPC Sponsor



This contest contains twelve problems. Good luck.

For problems that state “Your answer should have an absolute or relative error of less than 10^{-9} ”, your answer, x , will be compared to the correct answer, y . If $|x - y| < 10^{-9}$ or $\frac{|x - y|}{|y|} < 10^{-9}$, then your answer will be considered correct.

Definition 1

For problems that ask for a result modulo m :

If the correct answer to the problem is the integer b , then you should display the unique value a such that:

- $0 \leq a < m$
and
 - $(a - b)$ is a multiple of m .
-

Definition 2

A string $s_1 s_2 \dots s_n$ is lexicographically smaller than $t_1 t_2 \dots t_\ell$ if

- there exists $k \leq \min(n, \ell)$ such that $s_i = t_i$ for all $1 \leq i < k$ and $s_k < t_k$
or
 - $s_i = t_i$ for all $1 \leq i \leq \min(n, \ell)$ and $n < \ell$.
-

Definition 3

- Uppercase letters are the uppercase English letters ($A, B, \dots, Z$).
 - Lowercase letters are the lowercase English letters ($a, b, \dots, z$).
-

Definition 4

Unless otherwise specified, the distance between two points (x_0, y_0) and (x_1, y_1) is defined as its Euclidean distance:

$$\sqrt{(x_0 - x_1)^2 + (y_0 - y_1)^2}.$$

This page is intentionally left (almost) blank.

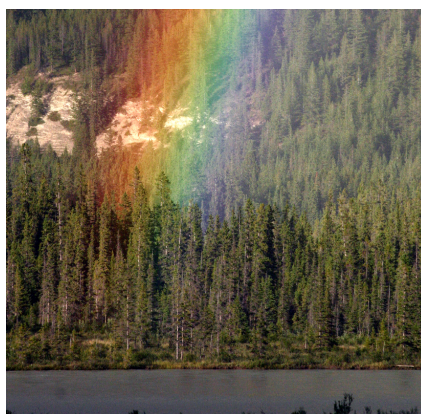
Problem A

Arr-Gee-Bee Rainbow

Time limit: 1 second

All input and output (in all problems) is given via `stdin` and `stdout`.

So you should use `input()` / `print()` in Python, `std::cin` / `std::cout` in C++, and `Scanner` / `System.out.println()` in Java to read and write. Good luck!



Source: Wing-Chi Poon, CC BY-SA 2.5 (adapted)

You have a string made of just the letters R (red), G (green), and B (blue). A string is a *rainbow* if all three colours are present, every R appears before every G, and every G appears before every B.

Is your string a rainbow?

Input

The input consists of a single line containing your string T . T contains at most 100 letters, and contains only the letters R, G, and B.

Output

Display YES if T is a rainbow. Otherwise, display NO.

Sample Input 1

RRRGGBBBBBBBB

Sample Output 1

YES

Sample Input 2

RGB

Sample Output 2

YES

Sample Input 3

RGBRBR

Sample Output 3

NO

Sample Input 4

GG

Sample Output 4

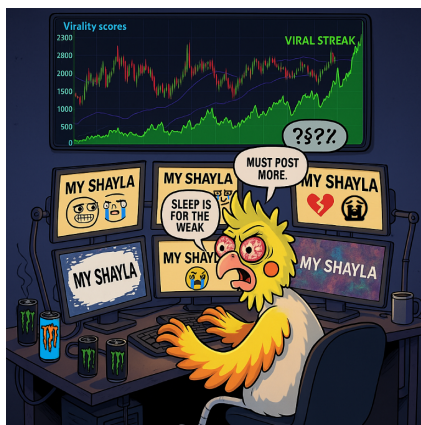
NO

This page is intentionally left (almost) blank.

Problem B

Broken Heart

Time limit: 8 seconds



Our extra epic and absolutely amazing friend (not a paid review) Tom, woke up one day and decided that she must become a part of the canon now! After deep reflection (and three energy drinks), she realized the best way to go viral was to post nonstop “MY SHAYLA 🥺🥺” memes on TicTac, GramGram, YouTuba, AOW, and every other platform in existence.

After many consecutive days of tireless meme posting, Tom collected detailed data on the virality of each of her posts. Each day is assigned a “virality score” measuring how well her content performed. Tom wants to know her longest viral streak. She defines a viral streak as a subarray of days whose arithmetic mean virality (meme mean) is greater than or equal to a threshold K . A subarray is a contiguous subsequence of days, and the arithmetic mean (arithmetic meme) is the sum of virality scores divided by the length of the subarray.

Input

The first line of input contains N ($1 \leq N \leq 10^6$) and K ($-10^9 \leq K \leq 10^9$), which are the number of days posted and the threshold respectively. The next line contains N integers $V_1, V_2, \dots, V_N$ ($-10^9 \leq V_i \leq 10^9$), which are the virality scores.

Output

Display the length of the longest viral streak. If there are no viral streaks, display 0.

Sample Input 1

```
5 3
4 5 6 7 8
```

Sample Output 1

```
5
```

Sample Input 2

```
6 4
1 2 10 8 1 2
```

Sample Output 2

```
6
```

Sample Input 3

```
4 1
-5 -4 -3 -2
```

Sample Output 3

```
0
```



Sample Input 4

7 5
9 1 1 1 1 10 10

Sample Output 4

4

Problem C

Combinatorial Dice

Time limit: 1 second



Source: an anonymous photographer (with permission)

You and your friends recently started playing *Dungeons & Dragons* with you as the DUNGEON MASTER. Each of your players has exactly one set of 6 dice: a $D4$, $D6$, $D8$, $D10$, $D12$, and $D20$ (where DX represents a fair die with X sides, numbered $1, 2, \dots, X$).

Critical moments of the game are decided by a roll of the dice to check if a player succeeds or fails in an action they are attempting. This is called a skill check.

For a particular skill check, you will instruct the player to roll some subset of their 6 dice and give them the target DC (difficulty class) which is a positive integer. If the sum of the rolled dice is at least the DC, then they succeed the skill check. Otherwise, they fail the skill check.

For example, you might instruct the player to roll their $D6$ (a 6-sided die) and their $D8$ (an 8-sided die) with a DC of 11. If they rolled a 4 on the $D6$ and a 7 on the $D8$, they will have succeeded the skill check (since $4 + 7 \geq 11$). If instead they rolled a 4 on the $D6$ and a 6 on the $D8$, they will have failed the skill check (since $4 + 6 < 11$).

You like to be precise in your skill checks and have already decided what the chance of success should be as a reduced fraction in the form n/d . What subset of dice should you instruct the player to roll and what DC should you set to make the chance of succeeding the skill check exactly n/d ?

Input

The first line of input contains two integers n ($1 \leq n \leq 10^9$) and d ($n \leq d \leq 10^9$) representing the desired chance of success as a reduced fraction n/d . It is guaranteed that there is at least one subset of the available dice and target DC that results in a chance of success of n/d .

Output

Display a string representing which dice the player should roll and the target DC ($1 \leq DC \leq 60$). The string representing which dice to roll should be a concatenation of the names of each die to roll ($D4$, $D6$, $D8$, $D10$, $D12$, and $D20$) in increasing order of the number of sides. For example, if the player should roll a $D6$ and a $D8$, the string should be $D6D8$. Each die rolled should be unique.

If there are multiple valid answers any will be considered correct.

Sample Input 1

5 24

Sample Output 1

D6D8 11

Sample Input 2

1 2

Sample Output 2

D12 7

Sample Input 3

19 20

Sample Output 3

D20 2



Sample Input 4

1 1

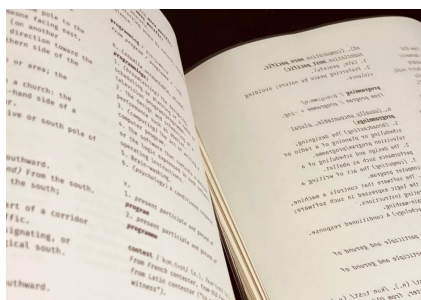
Sample Output 4

D4D6D8D10D12D20 6

Problem D

Dictionary Shuffle

Time limit: 8 seconds



Source: Hadyn Tang - using work of Wiktionary authors, CC BY-SA 4.0

You have a list of N words, which will be used to create a dictionary in lexicographic (alphabetical) order. However, you are allowed to reverse some (possibly all or none) of the words before sorting. This means there are many possible dictionaries, depending on which words you choose to reverse.

For each word in the list, determine the earliest and latest index (starting from 0) at which the word could appear in a dictionary, either in its original form or reversed, across all possible dictionaries.

Input

The first line contains a single integer N ($1 \leq N \leq 300\,000$), which is the number of words in the list.

Following are N lines, each containing a single string S_i which is the i th word in the list. Each string has at least one character and consists of only lowercase letters.

The sum of the lengths of all strings does not exceed 1 500 000. No two words are the same, and no word is the reverse of another word in the list. A word may, however, be the same as its own reverse.

The list is not necessarily in lexicographic order.

Output

Display N lines. The i th line should contain two integers, denoting the earliest and latest index at which the word S_i or its reverse could appear in a dictionary.

Sample Input 1

```
3
apple
banana
racecar
```

Sample Output 1

```
0 1
0 1
2 2
```

Sample Input 2

```
3
a
ab
abc
```

Sample Output 2

```
0 0
1 2
1 2
```

Sample Input 3

```
6
cat
dog
level
noon
stats
top
```

Sample Output 3

```
0 5
0 1
1 2
2 3
3 5
3 5
```

Problem E

Euclidean Extension

Time limit: 2 seconds



Source: Euclid, coloured woodcut, 1584, Public Domain (adapted)

A prime number has no integer factors greater than 1 other than itself. Euclid's proof that there are infinitely many primes relies on the fact that, given any set of primes, we can construct a number equal to 1 plus their product; this number has a prime factor not in the original set. For example, for the set $\{2, 17\}$, the number constructed is: $1 + (2 \times 17) = 35$, with prime factors 5 and 7.

Euclid's construction is a specific case of a more general form where a set S of primes is partitioned into two disjoint subsets ($S = S_1 \cup S_2$). Then, we compute the sum of the products of the two subsets ($\prod S_1 + \prod S_2$). Euclid's construction is the degenerate case where the pair of subsets consists of the empty set and the set itself. Note that the product of the empty set is 1 ($\prod \{\} = 1$) and the product of the set containing only the element x is x ($\prod \{x\} = x$).

For a given set S of primes, the numbers resulting from the possible partitions will be distinct. Define the resulting set as $E(S)$: the "extended Euclidean numbers" of S . Each number in $E(S)$ has the property that all of its prime factors are not in S .

Given a set S of primes, you are interested in generating new prime factors similar to Euclid's proof. Specifically, you want to find the set of all prime factors of numbers in the extended Euclidean numbers $E(S)$ of S .

For example, given the set $S = \{2, 3, 5\}$, its set of extended Euclidean numbers is $E(S) = \{11, 13, 17, 31\}$, where $11 = 5 + 2 \times 3$, $13 = 3 + 2 \times 5$, $17 = 2 + 3 \times 5$, and $31 = 1 + 2 \times 3 \times 5$.

In the previous example all the elements of $E(S)$ were prime, but this is not true in general. Say $S = \{2, 17\}$. Then $E(S) = \{35, 19\}$, where $35 = 1 + 2 \times 17$ and $19 = 2 + 17$. The resulting set of prime factors is $\{5, 7, 19\}$ (prime factors of 35 are 5 and 7, and 19 is prime).

Input

The input is a single line of integers representing the original set of primes. This line starts with N ($0 \leq N \leq 6$), which is the number of elements in the set. N distinct integers follow, each is a prime in the inclusive range from 2 to 127, and they are given in ascending order.

Output

Display a single line starting with an integer M , the number of distinct generated prime factors, followed by M integers describing the prime factors in ascending order.

Sample Input 1

0

Sample Output 1

1 2



Sample Input 2

2 2 17

Sample Output 2

3 5 7 19

Sample Input 3

3 2 3 5

Sample Output 3

4 11 13 17 31

Problem F

Firefighting Robots

Time limit: 5 seconds



Source: Milremrobotics, CC BY-SA 4.0

Firefreetown consists of just one long road with houses located at certain positions along it. Unfortunately, the town has no firefighters. The town wants to install special *firefighting robots* just in case a (very ironic) fire breaks out. Robots can only be installed at the same locations as houses because they need to be sheltered from the elements. The robots move at a speed of one metre per second.

Each house has a fire safety rating, indicating how many seconds it would take to burn down if it catches fire. A robot must be able to reach the house within this time to save it. A house with a robot installed in it is always safe.

What is the fewest robots the town needs to make sure no house will burn down if it catches fire? The town is only considering the case where a single house catches fire in their scenario.

Input

The first line of input contains the number of houses n ($1 \leq n \leq 300\,000$). The next n lines describe the houses. Each line contains two integers x_i ($0 \leq x_i \leq 10^9$) and s_i ($0 \leq s_i \leq 10^9$), the location of the i th house along the road (in metres from the start of the road) and the number of seconds it would take to burn down, respectively.

No two houses occupy the same location.

Output

Display one integer, the fewest robots the town needs.

Sample Input 1

```
3
0 2
2 1
4 2
```

Sample Output 1

```
1
```

Sample Input 2

```
2
5 0
1 4
```

Sample Output 2

```
1
```

Sample Input 3

```
4
1 10
3 1
5 2
10 5
```

Sample Output 3

```
2
```



Sample Input 4

```
2
1 8
10 8
```

Sample Output 4

```
2
```

Problem G

Grandma's House

Time limit: 1 second

Grandma's house is very cold. In fact, it is normally a frigid 0°C everywhere in her 10×10 living room. To fix this, she has placed some space heaters in the room. Each one is located in the middle of a 1×1 floor tile of her room.

A space heater increases the temperature of the tile it is on by 4°C ; it increases the temperature of the surrounding eight tiles by 3°C ; it increases the temperature of the next ring of 16 tiles by 2°C ; and it increases the temperature of the next ring of 24 tiles by 1°C .

	+1	+1	+1	+1	+1	+1	+1		
	+1	+2	+2	+2	+2	+2	+1		
	+1	+2	+3	+3	+3	+2	+1		
	+1	+2	+3	+4	+3	+2	+1		
	+1	+2	+3	+3	+3	+2	+1		
	+1	+2	+2	+2	+2	+2	+1		
	+1	+1	+1	+1	+1	+1	+1		

The change in temperature from a single space heater (located on the +4 tile).

A tile may have its temperature increased by more than one space heater.

Given the locations of the space heaters, what is the temperature of each tile?

Input

The input consists of 10 lines with 10 characters each, describing grandma's living room. If a tile has a space heater, the corresponding character is an asterisk (*). Otherwise, the tile has no space heater and the corresponding character is a dot (.).

Output

Display 10 lines with 10 integers each: the temperature of each tile in the room.

Sample Input 1

```
.....
.....
.....
.....
.....*.....
.....
.....
.....
.....
.....
```

Sample Output 1

```
0 0 0 0 0 0 0 0 0 0
0 1 1 1 1 1 1 1 0 0
0 1 2 2 2 2 2 1 0 0
0 1 2 3 3 3 2 1 0 0
0 1 2 3 4 3 2 1 0 0
0 1 2 3 3 3 2 1 0 0
0 1 2 2 2 2 2 1 0 0
0 1 1 1 1 1 1 1 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
```

Sample Input 2

```
.....
.....
.*.....
.....
.....
.....*.....
.....
.....
.....
.....
```

Sample Output 2

```
2 2 2 2 1 0 0 0 0 0
3 3 3 2 1 0 0 0 0 0
3 4 4 3 2 1 1 1 1 0
3 3 4 4 3 2 2 2 1 0
2 2 3 4 4 3 3 2 1 0
1 1 2 3 4 4 3 2 1 0
0 0 1 2 3 3 3 2 1 0
0 0 1 2 2 2 2 2 1 0
0 0 1 1 1 1 1 1 1 0
0 0 0 0 0 0 0 0 0 0
```

Sample Input 3

```
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
```

Sample Output 3

```
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
```

Sample Input 4

```
.....
.....
.....
.....
.....
.....*****
.....*****
.....*****
.....*****
.....*****
```

Sample Output 4

```
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
0 0 1 2 3 4 5 5 5 4
0 0 2 5 8 11 14 15 14 11
0 0 3 8 14 20 26 28 26 20
0 0 4 11 20 30 39 42 39 30
0 0 5 14 26 39 51 55 51 39
0 0 5 15 28 42 55 60 55 42
0 0 5 14 26 39 51 55 51 39
0 0 4 11 20 30 39 42 39 30
```

Problem H

Humanoid Robot Tango

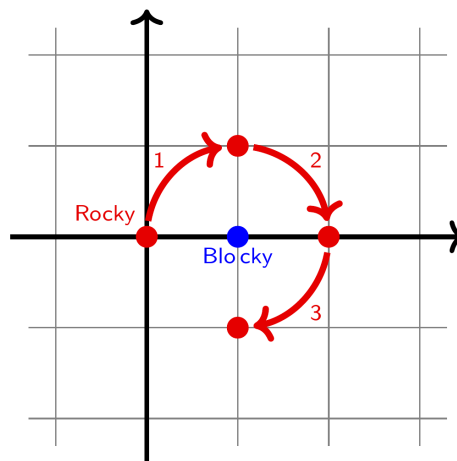
Time limit: 1 second

Rocky and Blocky are starting to choreograph a new dance—a Samba of Phlogiston Plus Cryogenics—which takes place on an infinite flat floor.

At the start of the dance, Rocky is at $(0, 0)$, and Blocky is at $(1, 0)$. Throughout the dance, one dancer is always designated as the active dancer and the other as the inactive dancer. Rocky is initially the active dancer.

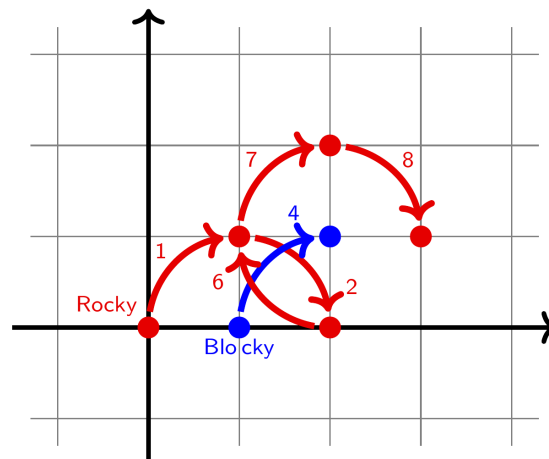
The dance consists of two different moves: slides (denoted by S) and bumps (denoted by B).

- In a slide, the active dancer rotates 90° clockwise around the inactive dancer at a fixed radius of 1 unit, while the inactive dancer remains in place. For example, if three consecutive slides occurred (which we would denote by SSS) Rocky's position would change as shown:



- In a bump, Rocky and Blocky swap their roles as the active and inactive dancer. That is, the person currently designated as the active dancer becomes the inactive dancer and vice-versa. They remain in the same positions during the bump.

By combining these types of moves, the dance becomes more complicated. The sequence of moves “slide, slide, bump, slide, bump, slide, slide, bump” (denoted SSBSBSSSB) would result in the following movements of the positions of the dancers:



Rocky is curious: at the end of the dance, where will he end up?

Input

The first line of input contains a single integer L ($1 \leq L \leq 10^6$), which is the number of moves in the dance. The second line contains a string with L characters which are either S or B, where the i th character is S if the i th move of the dance is a slide, or B if it is a bump.

Output

Display two integers, x and y , where (x, y) is Rocky's final position at the end of the dance.

Sample Input 1

```
3
SSS
```

Sample Output 1

```
1 -1
```

Sample Input 2

```
9
SSBSBSSSB
```

Sample Output 2

```
3 1
```

Problem I

I Get Extra Votes

Time limit: 4 seconds

The election is coming up. N candidates (numbered $1, 2, \dots, N$ by their pre-election rankings) are running to become the Supreme President and Really Cool Dude (SPRCD).



Source: Kurtkaiser, CC0 1.0 (adapted)

Each ballot has three different candidates listed, in order of preference. After all ballots are submitted (including any extra ones you might sneak in), the winner is determined by repeating the following process $N - 1$ times:

- Each ballot gives one vote to the highest-ranked candidate on it who has not yet been eliminated. If all candidates on the ballot have been eliminated, that ballot gives no votes.
- The candidate with the fewest votes is eliminated. If there are multiple candidates with the fewest votes, the one of those with the largest pre-election ranking is eliminated.

After this process, only one candidate remains. They are declared the winner (and immediately assume the office of SPRCD).

However, you are trying to destabilize the government by manipulating the voting system. Ballots are submitted electronically, and you have found a bug that allows you to submit exactly K identical copies of your ballot. You already know how everyone else is going to vote.

Determine all of the candidates that you can make win the election.

Input

The first line of input contains three integers N ($3 \leq N \leq 30$), which is the number of candidates, M ($1 \leq M \leq 30$), which is the number of other voters, and K ($1 \leq K \leq 10$), which is the number of identical ballots you submit.

The next M lines describe the other voters' ballots. Each of these lines contains three distinct integers v_1, v_2, v_3 ($1 \leq v_1, v_2, v_3 \leq N$), denoting that their ballot states that they prefer the candidate with pre-election ranking v_1 , then v_2 , and then v_3 .

Output

Display the pre-election rankings of all candidates that can win the election after you cast your K ballots. Display the numbers in ascending order.

Sample Input 1

```
3 3 1
1 2 3
1 2 3
3 2 1
```

Sample Output 1

```
1
```

Sample Input 2

```
3 3 1
1 2 3
3 2 1
3 2 1
```

Sample Output 2

```
1 3
```

Sample Input 3

```
4 3 1
3 4 1
2 3 4
2 4 3
```

Sample Output 3

```
1 2
```

Sample Input 4

```
4 3 10
4 2 1
4 2 1
4 2 1
```

Sample Output 4

```
1 2 3 4
```

Sample Input 5

```
9 4 2
3 1 4
1 5 9
2 6 5
3 5 8
```

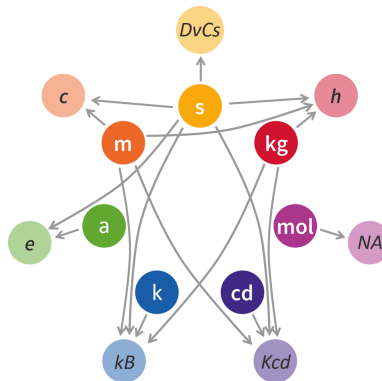
Sample Output 5

```
1 2 3 5 6 9
```

Problem J

Jeremy's Style Guide

Time limit: 1 second



Source: Emilio Pisanty, CC BY-SA 4.0 (adapted)

Jeremy loves formulas! In his Physics class, he gets a reference sheet filled with all sorts of funky formulas. It's all gibberish to you. Thankfully, Jeremy is eager to give you the crash course.

There are seven base units of measurement, known as the *SI units*¹: amperes (a), candelas (cd), kelvin (k), kilograms (kg), metres (m), moles (mol), and seconds (s).

The units of all *measurements* in Physics are a multiplication of zero or more SI units, each raised to a non-zero integer exponent. For example:

- velocity is measured in metres per second: $\text{m} \cdot \text{s}^{-1}$ (or $\frac{\text{m}}{\text{s}}$ in fractional form)
- electric charge is measured in second amperes: $\text{s} \cdot \text{a}$
- force is measured in kilograms times metres per second squared: $\text{kg} \cdot \text{m} \cdot \text{s}^{-2}$ (or $\frac{\text{kg} \cdot \text{m}}{\text{s}^2}$ in fractional form)
- 3 is a constant. It has no units.

Formulas also have units. Their units can be derived as a multiplication of the units of their measurements, each one raised to a non-zero integer exponent. For example:

- The formula for time is distance (m) multiplied by the reciprocal of velocity $((\text{m} \cdot \text{s}^{-1})^{-1})$. Its units would be $\text{m} \cdot (\text{m} \cdot \text{s}^{-1})^{-1} = \text{s}$.
- The formula for energy is mass (kg) multiplied by the speed of light squared $((\text{m} \cdot \text{s}^{-1})^2)$. Its units would be $\text{kg} \cdot (\text{m} \cdot \text{s}^{-1})^2 = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$ (or $\frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$ in fractional form).

You decide to write a program to help you determine the units of a formula from its measurements. You whip out your laptop and begin to cod–

“Wait!”

¹In the real world, amperes is A and kelvin is K. But for this problem, we will use lowercase letters instead.

Jeremy waves a units “style guide” in your face. In Jeremy’s class, the units for measurements and formulae:

- Must not omit the exponent when it is 1.
For example, the units for velocity should be written as $\text{m}^1 \cdot \text{s}^{-1}$.
- Must be in alphabetical order (a, cd, k, kg, m, mol, s).
For example, the units for electric charge should be written as $\text{a}^1 \cdot \text{s}^1$.
- Must never list the same SI unit more than once.

You tape the style guide to the side of your screen and continue to code.

Input

The first line contains the integer k ($1 \leq k \leq 10$), which is the number of measurements. The next k pairs of lines describe the measurements’ units (conforming to Jeremy’s style guide):

- The first line of the pair contains the name of the measurement as a string of uppercase and lowercase letters (at least 1 and at most 20 letters long).
- The second line of the pair describes the units. The line begins with an integer n ($0 \leq n \leq 7$), denoting the number of SI units. Then n pairs of values follow, describing the SI units:
 - The first value is a string denoting the SI unit: one of a, cd, k, kg, m, mol, or s.
 - The second value is the integer exponent e ($-5 \leq e \leq 5, e \neq 0$).

Then follows one last pair of lines describing the formula.

- The first line of the pair contains the name of the formula as a string of uppercase and lowercase letters (at least 1 and at most 20 letters long).
- The second line lists the measurements in the formula. The line begins with an integer m ($1 \leq m \leq 10$), denoting the number of measurements. Then m pairs of values follow, describing the measurements:
 - The first value is a string denoting the measurement name.
 - The second value is the integer exponent e ($-5 \leq e \leq 5, e \neq 0$).

Note that measurements may appear multiple times, and in any order in the formula description.

The measurement and formula names are distinct and are not any of the SI units (a, cd, k, kg, m, mol, or s).

Output

On the first line, display the name of the formula. On the second line, display the units of the formula according to Jeremy’s style guide, in the same format as the measurements from the input.

Sample Input 1

```
2
Mass
1 kg 1
SpeedOfLight
2 m 1 s -1
Energy
2 Mass 1 SpeedOfLight 2
```

Sample Output 1

```
Energy
3 kg 1 m 2 s -2
```

Sample Input 2

```
3
G
3 kg -1 m 3 s -2
Mass
1 kg 1
Distance
1 m 1
Force
4 G 1 Mass 1 Mass 1 Distance -2
```

Sample Output 2

```
Force
3 kg 1 m 1 s -2
```

Sample Input 3

```
5
Radius
1 m 1
Three
0
Pi
0
Four
0
Five
0
VolumeOfSphere
4 Four 1 Three -1 Pi 1 Radius 3
```

Sample Output 3

```
VolumeOfSphere
1 m 3
```

Sample Input 4

```
2
WaterVolume
2 kg 1 m -3
MaxWaterVolume
2 kg 1 m -3
Humidity
2 WaterVolume 1 MaxWaterVolume -1
```

Sample Output 4

```
Humidity
0
```

Sample Input 5

```
1
A
7 a -3 cd 1 k -4 kg 1 m 5 mol 1 s 1
AllUnits
1 A 5
```

Sample Output 5

```
AllUnits
7 a -15 cd 5 k -20 kg 5 m 25 mol 5 s 5
```

This page is intentionally left (almost) blank.

Problem K

Kayak is not a Palindrome

Time limit: 1 second

You have just finished training your pet drone to deliver snacks around your backyard. The drone can fly north, east, south, or west, one step at a time. After watching its paths for a while, you notice something satisfying: sometimes, its journey looks like it was perfectly undone. That is, it flew a certain path, then exactly traced its way back home step-by-step, but in reverse order. You call such a path a palindrome.

Formally, a flight path of length $2n$ is a *palindrome* if the path formed by the first n steps is exactly the same as the reverse of the path formed by the last n steps. For example, consider the path NESSWSNENNWS, where you start at $(0, 0)$.

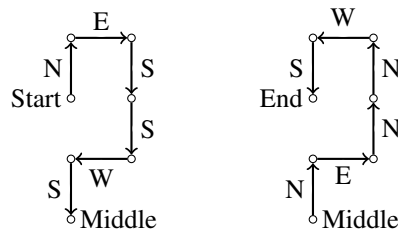


Figure K.1: First half starts at $(0, 0)$, then visits $(0, 1)$, then $(1, 1)$, then $(1, 0)$, then $(1, -1)$, then $(0, -1)$, then $(0, -2)$. Second half continues from $(0, -2)$, then visits $(0, -1)$, then $(1, -1)$, then $(1, 0)$, then $(1, 1)$, then $(0, 1)$, then $(0, 0)$.

Given the full flight path, determine whether it is a palindrome.

Input

The first line of input contains a single integer n ($1 \leq n \leq 100$), which indicates that the flight path is of length $2n$.

The second line of input contains a string of length $2n$, which is the flight path. The letters in the string are N (for north), E (for east), S (for south), and W (for west).

Output

If the flight path is a palindrome, display **Yes**. Otherwise, display **No**.

Sample Input 1

6
NESSWSNENNWS

Sample Output 1

Yes

Sample Input 2

2
NSEW

Sample Output 2

No

Sample Input 3

2
NEWS

Sample Output 3

Yes

Sample Input 4

4
NSNSNSNS

Sample Output 4

Yes

Problem L

Love Me

Time limit: 1 second

A lonely robot stands in a quiet flower field, trying to understand human emotions. From its vast database of scanned romance novels, it learns about an ancient ritual: Humans pluck petals from flowers, saying ‘they love me, they love me not,’ over and over.

But the robot misunderstood this ritual. Normally, you say one phrase per petal (*pluck*, ‘they love me’, *pluck*, ‘they love me not’, *pluck*, ‘they love me’, and so on) until the flower has no petals left. The phrase corresponding to the final petal is the *prophetic phrase*.

However, the robot thought they meant one word per petal (*pluck*, *they*, *pluck*, *love*, *pluck*, *me*, *pluck*, *they*, *pluck*, *love*, *pluck*, *me*, *pluck*, *not*, *pluck*, *they*, and so on). So the robot instead announces a *prophetic word* once it finishes plucking the flower.

For example, the prophetic word for a flower with 12 petals is *love* (they love me, they love me not, they love me, they **love**).

The prophetic word for flowers with up to 20 petals are shown in the table below.

# Petals	Prophetic Word	# Petals	Prophetic Word	# Petals	Prophetic Word	# Petals	Prophetic Word
1	they	2	love	3	me	4	they
5	love	6	me	7	not	8	they
9	love	10	me	11	they	12	love
13	me	14	not	15	they	16	love
17	me	18	they	19	love	20	me

Given the number of petals on the flower, what is the prophetic word?

Input

The input consists of a single line containing a single integer n ($1 \leq n \leq 10^{18}$), which is the number of petals on the flower.

Output

Display the prophetic word.

Sample Input 1

1

Sample Output 1

they

Sample Input 2

2

Sample Output 2

love

Sample Input 3

3

Sample Output 3

me

Sample Input 4

4

Sample Output 4

they

Sample Input 5

5

Sample Output 5

love

Sample Input 6

6

Sample Output 6

me



Sample Input 7

7	not
---	-----

Sample Output 7

Sample Input 8

1234567890123	they
---------------	------

Sample Output 8